

Lab 1: Forensic Disk Image Analysis

Using Autopsy and The Sleuth Kit (TSK)

Student: Almustapha Yusuf

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Evidence: Ch01InChap01.dd (FAT12 Floppy Disk Image)

ICDFA - International Certificate in Digital Forensics & Investigations

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1. Introduction and Objectives

This lab exercise focuses on the forensic analysis of a disk image using The Sleuth Kit (TSK), a suite of command-line tools for digital forensics investigations. The evidence provided is a raw disk image file named `Ch01InChap01.dd`, which is a forensic bit-for-bit copy of a 3.5-inch high-density floppy disk formatted with the FAT12 file system. The primary objectives of this lab include preserving the integrity of the original evidence through cryptographic hashing, analyzing the disk structure and file system metadata, identifying and recovering deleted files, searching unallocated disk space for relevant keywords, and documenting all findings in a manner consistent with forensic best practices and chain of custody requirements.

Throughout this exercise, we employ a systematic approach to digital forensics that mirrors real-world investigation workflows. The process begins with evidence identification and preservation, where we generate MD5 and SHA-256 hashes of the original disk image to establish a baseline for integrity verification. We then proceed to volume analysis using tools such as `img_stat` and `mmls` to understand the physical and logical structure of the disk. File system analysis follows, where `fsstat` and `fls` reveal the internal organization of the FAT12 file system, including the layout of the File Allocation Table, root directory, and data area. The critical phase of the investigation involves deleted file recovery using `istat` to examine inode metadata and `icat` to extract file contents from unallocated sectors. Finally, we conduct keyword searches across unallocated space and verify all recovered evidence through hash comparison.

The forensic tools used in this lab are part of The Sleuth Kit (TSK), an open-source forensic toolkit maintained by the Basis Technology Corporation. TSK provides a comprehensive set of utilities for analyzing disk volumes, file systems, and individual files. The specific tools employed include `img_stat` (image metadata), `mmls` (media management listing), `fsstat` (file system statistics), `fls` (file listing), `istat` (inode status), `icat` (inode content extraction), `blkcat` (block content extraction), `blkls` (unallocated block listing), and `tsk_recover` (automated file recovery). Each tool operates at a specific layer of the forensic analysis stack, from volume-level examination down to individual data block inspection, providing a thorough and methodical approach to evidence examination.

2. Lab Environment and Evidence Preparation

The forensic analysis was conducted on a Kali Linux system. Before beginning any analysis, it is essential to verify that all required forensic tools are installed and accessible, and to establish a proper workspace directory structure that separates evidence, recovered files, hash values, and documentation. This organizational discipline is a fundamental requirement of forensic investigations, ensuring that evidence handling is transparent, reproducible, and defensible in a court of law.

2.1 Tool Verification

The first step was to confirm that all nine Sleuth Kit utilities required for this lab are installed and available in the system PATH. The `which` command was used to locate each tool, confirming their presence in `/usr/bin/`. This verification step ensures that the analysis environment is properly configured before any evidence is examined, preventing potential issues mid-investigation due to missing dependencies. All required tools were confirmed to be present and functional on the Kali Linux system.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ which img_stat mmls fsstat fls istat icat blkcat blkls tsr_recover
/usr/bin/img_stat
/usr/bin/mmls
/usr/bin/fsstat
/usr/bin/fls
/usr/bin/istat
/usr/bin/icat
/usr/bin/blkcat
/usr/bin/blkls
/usr/bin/tsr_recover

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 1: Verification of Sleuth Kit tools - all 9 utilities confirmed in /usr/bin/

Tools Verified: img_stat, mmls, fsstat, fls, istat, icat, blkcat, blkls, tsr_recover

2.2 Evidence Acquisition and Hashing

The evidence file Ch01InChap01.dd was copied from the Desktop into a dedicated evidence/ directory within the forensic workspace at ~/Forensic_Lab3_Almustapha_Yusuf/. The file permissions were set to read/write for the owner only (rw-----), ensuring that no other users on the system could modify the evidence. The file was identified as a DOS/MBR boot sector image with a FAT12 file system, consistent with a standard 3.5-inch 1.44 MB floppy disk. Cryptographic hashes were then generated using both MD5 and SHA-256 algorithms to establish a chain of custody baseline. These hash values serve as digital fingerprints that can be used at any future point to verify that the evidence has not been altered, tampered with, or corrupted during the investigation process.

```
(kali@kali)-[~/Desktop]
$ cp Ch01InChap01.dd ~/Forensic_Lab3_Almustapha_Yusuf/evidence/

(kali@kali)-[~/Desktop]
$ cd ..

(kali@kali)-[~]
$ cd ~/Forensic_Lab3_Almustapha_Yusuf

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ ls -lh evidence/Ch01InChap01.dd
-rw----- 1 kali kali 1.5M Aug 29 10:16 evidence/Ch01InChap01.dd

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ file evidence/Ch01InChap01.dd
evidence/Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID "", 'k0nIHC' cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤32 MB)
, sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ md5sum evidence/Ch01InChap01.dd | tee hashes/source_md5.txt
a117773bcf1fc88ec0ab8e0a349fbbcb evidence/Ch01InChap01.dd

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ sha256sum evidence/Ch01InChap01.dd | tee hashes/source_sha256.txt
3ce8053e4f3d9c8ab98b3aad2480685efb8e4980d34297b83bd5a09b1a7b122 evidence/Ch01InChap01.dd
```

Figure 2: Evidence file identification, permission verification, and hash generation

File Name	Ch01InChap01.dd
File Size	1,474,560 bytes (1.44 MB)
File Type	DOS/MBR boot sector, FAT12
OEM ID	k0nIHC
Sectors	2880 (standard 3.5-inch floppy)
Sector Size	512 bytes
MD5 Hash	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA-256 Hash	3ce805364f3d9c8a980b3aad02480685ef8e4980d24297083bd5a09b1a7b0122

Table 1: Evidence file metadata and cryptographic hash values

3. Disk Image Analysis (Part B - The Sleuth Kit CLI)

3.1 Image Statistics (img_stat)

The `img_stat` utility provides essential metadata about the disk image file itself, including the image format, total size in bytes, and the sector size used by the original media. The output confirms that `Ch01InChap01.dd` is a raw (dd) image format, which means it contains a bit-for-bit copy of the original floppy disk without any compression or conversion. The total image size of 1,474,560 bytes corresponds exactly to the standard capacity of a 3.5-inch HD floppy disk (2880 sectors x 512 bytes per sector = 1,474,560 bytes). The 512-byte sector size is the standard for magnetic media including floppy disks and hard drives. Additionally, the `mmls` command was run to check for a partition table; however, no partition table was found, which is expected behavior for floppy disk media where the file system spans the entire disk without partitioning.

```
(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ mmls evidence/Ch01InChap01.dd

(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ img_stat evidence/Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size:    512

(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 3: `img_stat` output showing raw image format and `mmls` partition check

img_stat Results: Image Type = raw, Size = 1,474,560 bytes, Sector Size = 512 bytes

3.2 Partition Layout (mmls)

The mmls (Media Management Listing) utility was used to display the partition layout of the disk image. In the case of this evidence, mmls did not produce a traditional partition table output because floppy disk media typically do not contain a Master Boot Record (MBR) with partition entries. Instead, the entire disk is treated as a single volume containing one file system. This is consistent with the FAT12 file system identified by the file command earlier. On floppy disks, the boot sector at sector 0 contains the BIOS Parameter Block (BPB) directly, rather than a partition table followed by volume boot sectors as would be found on a hard disk.

3.3 Filesystem Statistics (fsstat)

The fsstat utility provides a comprehensive overview of the file system structure within the disk image. The output reveals detailed information about the FAT12 file system, including the OEM name (k0nIHC), volume layout in sectors, and content information such as cluster and sector sizes. The file system layout follows the standard FAT12 structure for a 1.44 MB floppy disk: the boot sector occupies sector 0, the first copy of the FAT spans sectors 1 through 9, the second FAT copy occupies sectors 10 through 18, the root directory is located in sectors 19 through 32, and the cluster area (data area for file contents) spans from sector 33 through sector 2879. Both the sector size and cluster size are 512 bytes, meaning each cluster corresponds to exactly one sector on this floppy disk. The total cluster range is 2 through 2848, providing a clear map of the data storage area.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ fsstat evidence/Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: *3*ml

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
-----
33-236 (204) -> EOF
285-311 (27) -> EOF
```

Figure 4: fsstat output showing complete FAT12 filesystem layout

Region	Sector Range	Sectors	Description
Boot Sector	0	1	BIOS Parameter Block and bootstrap code
FAT 0 (Primary)	1 - 9	9	First copy of File Allocation Table
FAT 1 (Backup)	10 - 18	9	Second copy of File Allocation Table
Root Directory	19 - 32	14	Root directory entries (224 max)

Region	Sector Range	Sectors	Description
Data Area	33 - 2879	2847	Cluster area for file storage

Table 2: FAT12 filesystem layout for Ch01InChap01.dd

3.4 File Listing (fls)

The fls utility lists the files and directories within a disk image, similar to the Unix ls command but operating on a disk image rather than a mounted file system. The output reveals six user files in the root directory, four of which are marked with an asterisk (*) indicating they have been deleted. The two remaining files without the asterisk are active (allocated) files that have not been deleted. The fls command was executed in three variations: standard listing, recursive listing (-r flag), and recursive listing with full paths (-r -p flags). All three produced identical output, confirming that this disk image contains a flat file system with no subdirectories. The virtual system entries (\$MBR, \$FAT1, \$FAT2, \$OrphanFiles) are TSK metadata pointers to structural components of the file system.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ fls evidence/Ch01InChap01.dd
r/r/r 5: Client Info.mdb
r/r/r * 8: Billing Letter.doc
r/r/r * 11: confirmation.txt
r/r/r 13: Income.xls
r/r/r * 15: letter1.txt
r/r/r * 17: Regrets.doc
v/v/v 45779: $MBR
v/v/v 45780: $FAT1
v/v/v 45781: $FAT2
v/v/v 45782: $OrphanFiles

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ fls -r evidence/Ch01InChap01.dd
r/r/r 5: Client Info.mdb
r/r/r * 8: Billing Letter.doc
r/r/r * 11: confirmation.txt
r/r/r 13: Income.xls
r/r/r * 15: letter1.txt
r/r/r * 17: Regrets.doc
v/v/v 45779: $MBR
v/v/v 45780: $FAT1
v/v/v 45781: $FAT2
v/v/v 45782: $OrphanFiles

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ fls -r -p evidence/Ch01InChap01.dd
r/r/r 5: Client Info.mdb
r/r/r * 8: Billing Letter.doc
r/r/r * 11: confirmation.txt
r/r/r 13: Income.xls
r/r/r * 15: letter1.txt
r/r/r * 17: Regrets.doc
v/v/v 45779: $MBR
v/v/v 45780: $FAT1
v/v/v 45781: $FAT2
v/v/v 45782: $OrphanFiles

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 5: fls output showing 6 files (4 deleted, 2 active) in root directory

File Name	Inode	Type	Status
Client Info.mdb	5	r/r (File)	Active (Allocated)
Billing Letter.doc	8	r/r * (File)	Deleted (*)
confirmation.txt	11	r/r * (File)	Deleted (*)
Income.xls	13	r/r (File)	Active (Allocated)
letter1.txt	15	r/r * (File)	Deleted (*)
Regrets.doc	17	r/r * (File)	Deleted (*)

Table 3: File listing from fls - allocated and deleted files identified

4. Deleted File Analysis and Recovery

4.1 Inode Metadata Examination (istat)

The `istat` utility displays detailed metadata for a specific file identified by its inode number. This tool is critical for forensic investigations because it reveals information about deleted files that would otherwise be inaccessible through normal file system operations. For each of the four deleted files, `istat` provided the allocation status (confirming they are Not Allocated), file attributes, logical file size, the short file name as stored in the directory entry, three timestamps (written, accessed, and created), and the list of sectors where the file data resides. A key observation is that all four files were created (had their directory entries modified) within a four-second window on December 9, 2005, between 06:59:05 and 06:59:09, suggesting they were deleted as part of a single user action or session. Despite being deleted, the data sectors remain intact and contiguous on the disk, making recovery straightforward.

```
(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ istat evidence/Ch01InChap01.dd 8
Directory Entry: 8
Not Allocated
File Attributes: File, Archive
Size: 24064
Name: _ILLIN~1.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:28 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:05 (EST)

Sectors:
237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283

(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ istat evidence/Ch01InChap01.dd 11
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written:      2005-12-09 06:52:58 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:06 (EST)

Sectors:
284
```

Figure 6: `istat` output for inodes 8 (Billing Letter.doc) and 11 (confirmation.txt)

```
(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ istat evidence/Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:09 (EST)

Sectors:
312

(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ istat evidence/Ch01InChap01.dd 17
Directory Entry: 17
Not Allocated
File Attributes: File, Archive
Size: 23552
Name: _EGRETS.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:52 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:09 (EST)

Sectors:
313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328
329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344
345 346 347 348 349 350 351 352
353 354 355 356 357 358

(kali㉿kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 7: `istat` output for inodes 15 (letter1.txt) and 17 (Regrets.doc)

File	Inode	Size (bytes)	Status	Written	Created	Data Sectors
Billing Letter.doc	8	24,064	Not Allocated	2005-12-09 06:50:28	2005-12-09 06:59:05	237-283 (47 sectors)
confirmation.txt	11	227	Not Allocated	2005-12-09 06:52:58	2005-12-09 06:59:06	284 (1 sector)
letter1.txt	15	121	Not Allocated	2005-12-09 06:51:50	2005-12-09 06:59:09	312 (1 sector)
Regrets.doc	17	23,552	Not Allocated	2005-12-09 06:50:52	2005-12-09 06:59:09	313-358 (46 sectors)

Table 4: istat metadata for all four deleted files

4.2 File Recovery with icat

The icat utility extracts the contents of a file from a disk image by reading the data from the sectors listed in the file inode. Unlike normal file copy operations that rely on the file system directory structure, icat works directly with inode numbers, making it ideal for recovering deleted files whose directory entries have been marked as free but whose data clusters have not yet been overwritten. Four icat commands were executed, one for each deleted file inode (8, 11, 15, and 17), with output redirected to appropriately named files in the recovered/ directory. All four commands executed successfully without errors, confirming that the deleted file data was still intact and recoverable from the disk image.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 8 > recovered/Billing_Letter.doc

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 11 > recovered/confirmation.txt

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 15 > recovered/letter1.txt

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 17 > recovered/Regrets.doc
```

Figure 8: icat commands recovering all four deleted files by inode number

4.3 Recovered File Contents

After recovery, the text files were examined using the cat command to verify their contents, and all recovered files were analyzed using the file command to confirm their types and sizes. The two text files yielded particularly significant forensic evidence. The confirmation.txt file contains FTP login credentials belonging to a user named Laura Roper, including a username (laura.roper), password, and server address (ftp.acmeserver.com). The letter1.txt file contains a brief message from George to Earl regarding a meeting scheduled for August 18th. The file command confirmed that both .doc files are legitimate Microsoft Word 9.0 documents

authored by Amelia Phillips, with creation dates in November 2002 and last modified dates of December 9, 2005. All file sizes matched exactly with the sizes reported by istat, confirming complete and accurate recovery.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
└─$ cat recovered/confirmation.txt
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
└─$ cat recovered/letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
└─$ file recovered/*
recovered/Billing_Letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
recovered/confirmation.txt: ASCII text, with CRLF line terminators
recovered/letter1.txt: ASCII text, with CRLF line terminators
recovered/Regrets.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 02 November 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 02:00, Create Time/Date: Sat Nov 23 03:12:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 58, Number of Characters: 336, Security: 0
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
└─$ ls -la recovered/
total 64
drwxrwxr-x 2 kali kali 4096 Aug 29 10:40 .
drwxrwxr-x 8 kali kali 4096 Aug 29 10:13 ..
-rw-rw-r-- 1 kali kali 24064 Aug 29 10:39 Billing_Letter.doc
-rw-rw-r-- 1 kali kali 227 Aug 29 10:40 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Aug 29 10:40 letter1.txt
-rw-rw-r-- 1 kali kali 23552 Aug 29 10:40 Regrets.doc
```

Figure 9: Verification of recovered files - contents, file types, and directory listing

Contents of recovered/confirmation.txt:

Laura, The domain has been registered and you can upload your files by going to this ftp address using your browser. ftp.acmeserver.com login id: laura.roper password: 342rroi9 Thank you for your business George

Contents of recovered/letter1.txt:

Earl, We need to meet on the 18th of August to confirm the work I am doing for you. Please contact me ASAP. George

File	Size	Type	Key Metadata
Billing Letter.doc	24,064 B	MS Word 9.0 (OLE2)	Author: Amelia Phillips, 118 words, Created: 2002-11-23
confirmation.txt	227 B	ASCII text (CRLF)	Contains FTP credentials for laura.roper
letter1.txt	121 B	ASCII text (CRLF)	Message from George to Earl about Aug 18 meeting
Regrets.doc	23,552 B	MS Word 9.0 (OLE2)	Author: Amelia Phillips, 58 words, Created: 2002-11-23

Table 5: Recovered deleted files - types, sizes, and key metadata

4.4 Active File Extraction

In addition to recovering deleted files, the two active (allocated) files were also extracted from the disk image using icat with their respective inode numbers. Inode 5 corresponds to Client Info.mdb, which was identified by the file command as a Microsoft Access Database with a size of 104,448 bytes. Inode 13 corresponds to Income.xls, identified as a Microsoft Excel spreadsheet with a size of 13,824 bytes. Both files were authored by Amelia Phillips according to their OLE metadata, with creation dates in November 2002 and last modification dates on December 9, 2005. These active files were placed in a separate active/ directory to maintain a

clear distinction between recovered deleted evidence and allocated files that were still present on the disk when the image was acquired.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ mkdir -p active

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 5 > active/Client_Info.mdb

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ icat evidence/Ch01InChap01.dd 13 > active/Income.xls

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ file active/*
active/Client_Info.mdb: Microsoft Access Database
active/Income.xls:      Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved
By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18
2005, Security: 0

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ ls -la active/
total 128
drwxrwxr-x 2 kali kali  4096 Aug 29 10:48 .
drwxrwxr-x 9 kali kali  4096 Aug 29 10:48 ..
-rw-rw-r-- 1 kali kali 104448 Aug 29 10:48 Client_Info.mdb
-rw-rw-r-- 1 kali kali  13824 Aug 29 10:48 Income.xls

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 10: Active file extraction and verification - Client Info.mdb and Income.xls

File	Inode	Size	Type	Author
Client Info.mdb	5	104,448 bytes	Microsoft Access Database	N/A (OLE metadata)
Income.xls	13	13,824 bytes	MS Excel (OLE2)	Amelia Phillips

Table 6: Active (allocated) files extracted from the disk image

5. Unallocated Space Analysis

5.1 Keyword Search (blkls + strings + grep)

The blkls utility extracts data from unallocated (slack and free) space within a disk image. By piping its output through the strings utility (which extracts printable ASCII strings from binary data) and then through grep with carefully chosen keywords, we can identify remnants of deleted files that persist in unallocated space. The keywords selected for this search were based on information discovered during the initial file recovery phase: names of individuals (laura, george, earl), the organization name (acme), the protocol (ftp), and the sensitive term (password). The search yielded significant results, confirming that data from deleted files persists in unallocated disk space even after the files have been removed from the directory structure. Notably, the FTP credentials, email addresses, and website URLs from the deleted documents were all recoverable from unallocated space, along with full names of the individuals involved in the correspondence found on the disk.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ blkls evidence/Ch01InChap01.dd | strings | grep -i "password\\|laura\\|acme\\|ftp\\|george\\|earl"
Laura Roper
Dear Laura,
  HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
George Montgomery
  HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
Laura,
files by going to this ftp address using your browser.
ftp.acmeserver.com
login id: laura.roper
password: 342rroi9
George
Earl,
George
  HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
```

Figure 11: Keyword search in unallocated space using blkls, strings, and grep

Key findings from unallocated space search:

Keyword / Artifact	Significance
laura.roper	FTP username found in deleted confirmation.txt
3422rroi9	FTP password found in unallocated space
ftp.acmeserver.com	FTP server address from deleted file
george.montgomery@nowhere.com	Email address from deleted .doc files
www.lauras_stuff.com	Website URL found in deleted document
Laura Roper / George Montgomery / Earl	Full names of individuals in correspondence

Table 7: Key artifacts recovered from unallocated space keyword search

5.2 Boot Sector Examination (blkcat)

The blkcat utility extracts the raw contents of a specific sector (block) from a disk image. Sector 0 was examined using blkcat piped through xxd (a hex dump utility) to inspect the boot sector of the FAT12 file system. The hex dump reveals the standard FAT12 boot sector structure: the first byte (0xEB) is a jump instruction, followed by the OEM identifier string, the BIOS Parameter Block (BPB) starting at offset 3, and bootstrap code. The OEM ID confirmed as the value seen earlier in the file command output. This low-level examination demonstrates the ability to inspect raw disk structures at the byte level, which is essential for understanding file system internals and detecting anomalies that might indicate evidence tampering or hidden data.

```

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ blkcat evidence/Ch01InChap01.dd 0 | xxd | head -20
00000000: eb34 902c 276b 306e 4948 4300 0201 0100  .4., 'k0nIHC.....
00000010: 02e0 0040 0bf0 0900 1200 0200 0000 0000  ... @.....
00000020: 0000 0000 0000 0000 0000 0000 0000 0012  ....
00000030: 0000 0000 0100 fa33 c08e d0bc 007c 1607  ....3.....l...
00000040: bb78 0036 c537 1e56 1653 bf2b 7cb9 0b00  .x.6.7.V.S.+l...
00000050: fcac 2680 3d00 7403 268a 05aa 8ac4 e2f1  ...G...+l...rg..
00000060: 061f 8947 02c7 072b 7cfb cd13 7267 a010  |..6.l...l...l?
00000070: 7c98 f726 167c 0306 1c7c 0306 0e7c a33f  |.7l. ...6.l...l.
00000080: 7ca3 377c b820 00f7 2611 7c8b 1e0b 7c03  .H....7l....?l..
00000090: c348 f7f3 0106 377c bb00 05a1 3f7c e89f  .7l. ...6.l...l.
000000a0: 00b8 0102 e8b3 0072 198b fbb9 0b00 bed6  }..u... ..}.
000000b0: 7df3 a675 0d8d 7f20 bee1 7db9 0b00 f3a6  }.....r.....
000000c0: 7418 be77 7de8 6a00 32e4 cd16 5e1f 8f04  t..w}.j.2...^...
000000d0: 8f44 02cd 19be c07d ebeb a11c 0533 d2f7  .D.....}.3..
000000e0: 360b 7cfe c0a2 3c7c a137 7ca3 3d7c bb00  6.l...<.7l.=...
000000f0: 07a1 377c e849 00a1 187c 2a06 3b7c 4038  ..7l.I...l*.;l@8
00000100: 063c 7c73 03a0 3c7c 50e8 4e00 5872 c628  .<S.. <P.N.Xr.(
00000110: 063c 7c74 0c01 0637 7cf7 260b 7c03 d8eb  .<t...7l.6.l...
00000120: d08a 2e15 7c8a 16fd 7d8b 1e3d 7cea 0000  ....l...}.=l...
00000130: 7000 ac0a c074 22b4 0ebb 0700 cd10 ebf2  p....t".....

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ tsk_recover -e evidence/Ch01InChap01.dd recovered_all/
Files Recovered: 6

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ ls -la recovered_all/
total 184
drwxrwxr-x 2 kali kali 4096 Aug 29 10:54 .
drwxrwxr-x 10 kali kali 4096 Aug 29 10:54 ..
-rw-rw-r-- 1 kali kali 24064 Aug 29 10:54 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 104448 Aug 29 10:54 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 227 Aug 29 10:54 confirmation.txt
-rw-rw-r-- 1 kali kali 13824 Aug 29 10:54 Income.xls
-rw-rw-r-- 1 kali kali 121 Aug 29 10:54 letter1.txt
-rw-rw-r-- 1 kali kali 23552 Aug 29 10:54 Regrets.doc

```

Figure 12: Hex dump of boot sector (sector 0) via blkcat and tsk_recover output

6. Automated Recovery and Verification

6.1 Automated Recovery with tsk_recover

The tsk_recover utility provides automated file recovery capabilities, extracting all files (both allocated and deleted) from a disk image into a specified output directory. The -e flag was used to include all file types during recovery. The tool successfully recovered all six files that were previously identified through the manual fls listing: the two active files (Client Info.mdb and Income.xls) and the four deleted files (Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc). The file sizes in the recovered_all/ directory match exactly with those obtained through the manual icat extraction, confirming that tsk_recover produces identical results. This automated approach serves as a valuable validation tool and would be the preferred method in a time-constrained investigation where comprehensive recovery of all files is required quickly.

6.2 Hash Verification of Recovered Files

To complete the chain of custody and ensure the integrity of the recovered evidence, MD5 hashes were computed for the recovered text files. These hash values can be used in future proceedings to demonstrate that the recovered files have not been modified since their extraction from the disk image. The MD5 hash of confirmation.txt is 18e391549e4a48bc990b264f590fb33b, and the MD5 hash of letter1.txt is 49f681040660a0914c015e86f3151237. These values should be recorded in the case documentation and cross-referenced if the files are presented as evidence. The final directory structure of the forensic workspace was documented using the tree command, showing nine directories and fifteen files organized into evidence, hashes, recovered, recovered_all, active, screenshots, tsk_outputs, and autopsy_report directories.

```
(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ md5sum recovered/confirmation.txt recovered/letter1.txt
18e391549e4a8bc990b264f590fb33bb  recovered/confirmation.txt
49f681040660a091a4c015e86f3151237  recovered/letter1.txt

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$ tree ~/Forensic_Lab3_Almustapha_Yusuf/
/home/kali/Forensic_Lab3_Almustapha_Yusuf/
├── active
│   ├── Client_Info.mdb
│   └── Income.xls
├── autopsy_report
├── evidence
│   └── Ch01InChap01.dd
├── hashes
│   ├── source_md5.txt
│   └── source_sha256.txt
├── recovered
│   ├── Billing_Letter.doc
│   ├── confirmation.txt
│   ├── letter1.txt
│   └── Regrets.doc
├── recovered_all
│   ├── Billing Letter.doc
│   ├── Client Info.mdb
│   ├── confirmation.txt
│   ├── Income.xls
│   ├── letter1.txt
│   └── Regrets.doc
├── screenshots
└── tsk_outputs

9 directories, 15 files

(kali@kali)-[~/Forensic_Lab3_Almustapha_Yusuf]
$
```

Figure 13: Hash verification of recovered files and final workspace directory structure

File	MD5 Hash
confirmation.txt	18e391549e4a48bc990b264f590fb33b
letter1.txt	49f681040660a0914c015e86f3151237

Table 8: MD5 hash values for chain of custody verification

7. Summary of Forensic Findings

The forensic analysis of the Ch01InChap01.dd disk image yielded significant findings that would be relevant to an investigation. The disk image contained six files in total: two active (allocated) files and four deleted files. All deleted files were successfully recovered with their original content intact, as the data sectors had not been overwritten since deletion. The analysis revealed a network of individuals engaged in business correspondence and file sharing via FTP, with the primary actors being Laura Roper, George Montgomery, and an individual known only as Earl. The most significant forensic artifact discovered was a set of plaintext FTP login credentials (username: laura.ropер, password: 342rroi9) for the server ftp.acmeserver.com, found in a deleted confirmation letter. This credential information was corroborated by the keyword search of unallocated space, which also recovered email addresses, website URLs, and additional context from deleted Word documents.

Evidence Category	Details
Disk Image	FAT12 floppy (1.44 MB), 2880 sectors, raw dd format
Total Files Found	6 (2 active + 4 deleted)
Files Successfully Recovered	6 of 6 (100%)
Deleted File Contents Intact	Yes - all data sectors unmodified since deletion
FTP Credentials Found	laura.ropер / 342rroi9 @ ftp.acmeserver.com
Individuals Identified	Laura Roper, George Montgomery, Earl, Amelia Phillips (doc author)
Email Address	george.montgomery@nowhere.com

Evidence Category	Details
Website	www.lauras_stuff.com
Document Author	Amelia Phillips (MS Word 9.0, Office 2000)
File Origin	Windows system (CRLF line terminators, OLE2 format)
Timeline	Files created Nov 2002, last modified Dec 9, 2005

Table 9: Comprehensive summary of forensic findings

8. Conclusion

This laboratory exercise successfully demonstrated the systematic forensic analysis of a disk image using The Sleuth Kit (TSK) command-line utilities. The investigation followed a methodical workflow that began with evidence preservation through cryptographic hashing, progressed through volume and file system analysis, and culminated in the successful recovery of deleted files and the extraction of intelligence from unallocated disk space. The FAT12 file system on the floppy disk image proved to be well-suited for forensic analysis due to its simple structure and the fact that the deleted file data had not been overwritten. All four deleted files were recovered with complete integrity, as confirmed by matching file sizes between the istat metadata and the recovered files.

The exercise highlighted several important principles of digital forensics. First, the importance of cryptographic hashing for maintaining chain of custody was demonstrated by generating MD5 and SHA-256 hashes of the evidence before and after analysis. Second, the value of layered analysis was shown by using multiple TSK tools at different levels of the storage hierarchy, from image-level (img_stat) through file-system-level (fsstat, fls) to file-level (istat, icat) and block-level (blkcat, blkls) examination. Third, the persistence of data in unallocated space was confirmed through the keyword search, which recovered sensitive information including login credentials that had been deleted from the active file system. These findings underscore the fundamental forensic principle that deletion does not erase data, and that proper forensic tools and techniques can recover evidence even after it has been removed from the file system directory structure.

The tools and techniques practiced in this lab are directly applicable to real-world digital forensic investigations. TSK utilities are widely used in law enforcement, military, and private-sector forensic examinations. The workflow demonstrated here, from evidence acquisition through analysis to reporting, reflects industry best practices as documented by organizations such as NIST (National Institute of Standards and Technology) and SWGDE (Scientific Working Group on Digital Evidence). The ability to recover deleted files, extract metadata, search unallocated space, and maintain a defensible chain of custody are core competencies for any digital forensics practitioner.